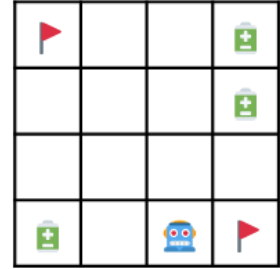


Q1)

Consider a delivery drone moving on a $N \times N$ grid.

- At each time step, the drone must move to an adjacent square (up, down, left, right). All actions cost 1.
- There are k fixed destination squares on the grid. The drone needs to visit all k destinations at least once.
- The drone starts with a battery charge of C units, and must always have at least 1 unit of charge at every time step. The drone loses 1 unit of charge per action.
- There are p power cells on the grid (with fixed locations). When the drone visits a power cell, its battery charge resets to C units, and the power cell cannot be used again.



Sample grid with $N = 4$,
 $k = 2$ destinations ,
 $p = 3$ power cells ,
 and drone .

What is the maximum branching factor for this problem?

What is the size of the smallest state space representation for this problem?

Algorithms that are guaranteed to find the lowest-cost solution?

☐ A DFS tree search

☐ E UCS tree search

☐ B DFS graph search

☐ F UCS graph search

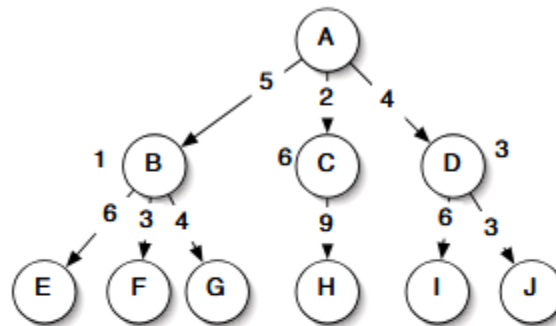
☐ C BFS tree search

☐ G None of the above

☐ D BFS graph search

Q2)

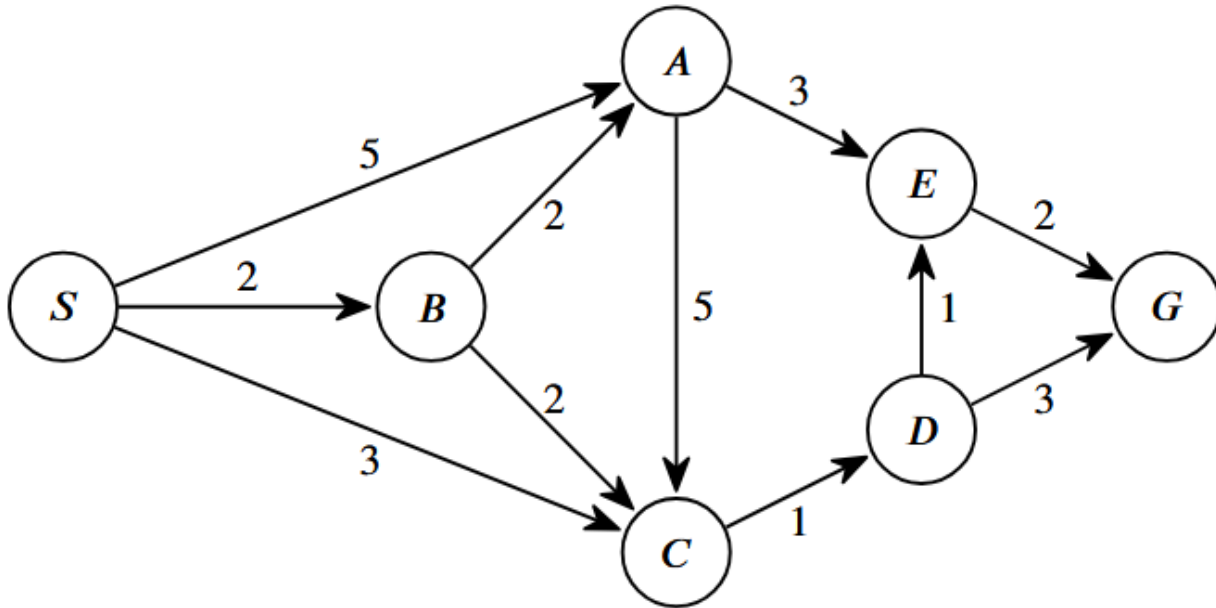
Consider the tree shown below. The numbers on the arcs are the arc lengths; the numbers near states B, C, and D are the heuristic estimates; all other states have a heuristic estimate of 0.



Assume that the children of a node are expanded in alphabetical order when no other order is specified by the search, and that the goal is state *J*. No visited or expanded lists are used. What order would the states be expanded by each type of search. Write only the sequence of states expanded by each search.

Search Type	List of states
Breadth First	
Depth First	
Progressive Deepening Search	
Best-First Search	
A* Search	

Q3)



What will be the path returned by **DFS Graph Search**?

What will be the path returned by **BFS Tree Search**?

In **BFS Graph Search**, how many times does the successor function get called on state *C*?

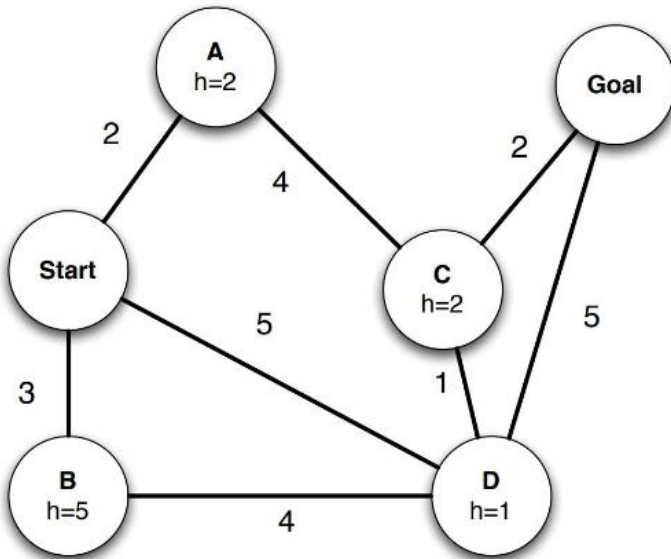
Q4)

TRUE / FALSE

1. UCS graph search is guaranteed to find the optimal solution.
2. You design a heuristic function h such that for all states s , $h(s) > 0$. If you run greedy graph search on the problem using the heuristic, it is guaranteed to get the optimal solution.
3. Suppose both UCS graph search and greedy graph search return the same optimal path for a problem. If you use the heuristic in 2, it is guaranteed that A* graph search will also return the optimal path.
4. If a heuristic is consistent, it is also admissible. True
- 5.

Q5)

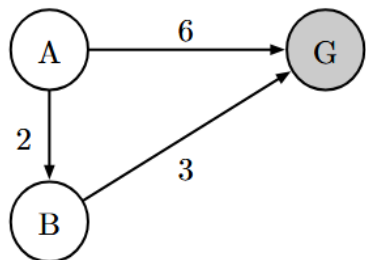
For each of the following graph search strategies, work out the order in which states are expanded, as well as the path returned by graph search. In all cases, assume ties resolve in such a way that states with earlier alphabetical order are expanded first. The start and goal state are S and G, respectively. Remember that in graph search, a state is expanded only once.



- Depth-first search.
- Breadth-first search.
- Uniform cost search.
- Greedy search with the heuristic h shown on the graph.
- A* search with the same heuristic.

Q6)

For the following questions, consider the search problem shown on the left. It has only three states, and three directed edges. A is the start node and G is the goal node. To the right, four different heuristic functions are defined, numbered I through IV.



	$h(A)$	$h(B)$	$h(G)$
I	4	1	0
II	5	4	0
III	4	3	0
IV	5	2	0

For each heuristic function, circle whether it is admissible and whether it is consistent with respect to the search problem given above.

	Admissible?		Consistent?	
I	Yes	No	Yes	No
II	Yes	No	Yes	No
III	Yes	No	Yes	No
IV	Yes	No	Yes	No

Circle all true statements among the following.

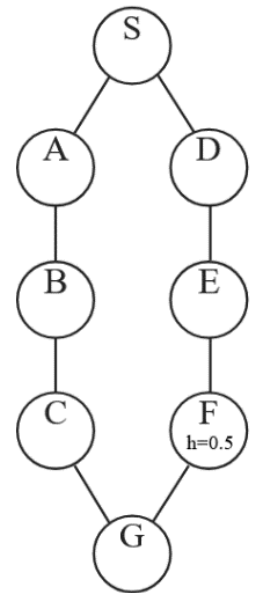
1. Heuristic function III dominates IV.
2. Heuristic function IV dominates III.
3. Heuristic functions III and IV have no dominance relationship.
4. Heuristic function I dominates IV.
5. Heuristic function IV dominates I.
6. Heuristic functions I and IV have no dominance relationship.

Q7)

For the following question parts, all edges in the graphs discussed have cost 1.

Suppose that you are designing a heuristic h for the graph on the right. You are told that $h(F) = 0.5$, but given no other information. What ranges of values are possible for $h(D)$ if the following conditions must hold? Your answer should be a range, e.g. $2 \leq h(D) < 10$. You may assume that h is nonnegative.

- i. h must be admissible
- ii. h must be admissible and consistent



Q8)

For each of the following situations, pick the search that is most appropriate (be specific about visited and expanded list). Give a one sentence reason why you picked it. If you write a paragraph, we will not read it.

1. We have a very large search space with a large branching factor and with possibly infinite paths. We have no heuristic. We want to find paths to the goal with minimum numbers of state.
2. We have a space with a manageable number of states but lots of cycles in the state graph. We have links of varying costs but no heuristic and we want to find shortest paths.
3. Our search space is a tree of fixed depth and all the goals are the leaves of the tree. We have a heuristic and we want to find any goal as quickly as possible.
4. We have a space with a manageable number of states but lots of cycles in the state graph. We have links of varying costs and an admissible heuristic and we want to find shortest paths.